

COMPUTER NETWORK

Title: Secure File Converter using TCP and SSL

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Abstract

This project presents a secure client-server based file conversion system using TCP sockets and SSL encryption. It enables users to convert files between multiple formats such as TXT, PDF, DOCX, CSV, PNG, JPG, and Binary. The system ensures secure transmission, efficient processing using SJF scheduling, and reliable output delivery.

1. Introduction

With the increasing need for file compatibility, file conversion systems have become essential. This project focuses on secure file conversion over networks using encryption and efficient scheduling techniques.

2. Objectives

- 1.To implement secure file transfer using SSL.
- 2.To support multiple file format conversions.
- 3.To implement SJF scheduling using priority queue.
- 4.To enable binary (0s and 1s) conversion for all file types.

3. System Architecture

The system follows a client-server architecture. The client provides a GUI for user interaction, while

the server handles file processing and conversion.

4. Working

- 1.User selects file and format in client GUI.
- 2.Client sends file securely to server using SSL.
- 3.Server receives file and places it in SJF queue.
- 4.File is processed and converted.
- 5.Converted file is sent back to client.

5. Technologies Used

- 1.Python Programming
- 2.TCP Sockets
- 3.SSL Encryption
- 4.Tkinter GUI
- 5.FPDF, PyPDF2, python-docx, Pillow

6. Binary Conversion

The system converts files into binary format using 8-bit representation for each byte. This enables universal conversion and reconstruction of files.

7. Advantages

- 1.Secure data transmission
- 2.Supports multiple file formats
- 3.Efficient processing using SJF
- 4.User-friendly GUI

8. Conclusion

The project successfully demonstrates secure and efficient file conversion using networking and encryption techniques. It provides a flexible solution for modern file compatibility needs.